Movie Rental Business Analysis

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Data: Sakila DVD rental database (MySQL)
Tools: SQL subqueries and an Excel dashboard
Period covered: May to August 2005, plus February 2006

1. Objective

- Understand how a two-store DVD rental business earns its money.
- Find the best customers, the best films and categories, and the dead stock.
- Compare the two stores and the two staff members.
- Suggest practical actions for stock, staff and promotions.

2. Data and approach

- Data: 16,044 rentals, 599 customers, 1,000 films, 16 film categories.
- SQL: seven queries, each built around a subquery.
    - Customers who spend above the average
    - Top 10 most rented films
    - Films never rented
    - Customers above their own store's average
    - Revenue by category
    - Staff workload
    - Customer lifetime value ranking
- Excel: loaded the results, built PivotTables, and made a one-page dashboard with filters for store, category and rating.
- Checks: compare every Excel total with the database.
    - Row counts, total revenue and all query results matched.
    - No duplicate or missing values were found.

3. Key findings

Revenue and timing

- Total revenue is 67,406.56 from 16,044 rentals, about 4.20 per rental.
- July and August together bring in 77.8% of all revenue.
- There are no rentals from September 2005 to January 2006.

Month	Revenue	Rentals
May 2005	4,823.44	1,156
Jun 2005	9,629.89	2,311

Jul 2005	28,368.91	6,709
Aug 2005	24,070.14	5,686
Feb 2006	514.18	182

- *Customers*
- The average customer spent 112.53 in total.
- 285 of 599 customers (47.6%) spent more than that average.
- The top 10 customers spent roughly 1.5 to 2 times the average.
- 296 customers rented more than the average customer of their own store.

Films

- Most rented: Bucket Brotherhood, with 34 rentals.
- The top 10 films were each rented 31 to 34 times, about twice the average of 16.
- 42 films (4.2% of the catalogue) were never rented.

Categories

- Sports is first with 5,314.21 (7.88% of revenue).
- Sci-Fi is second (4,756.98), and Animation is third (4,656.30).
- Music is lowest with 3,417.72 (5.07%).
- The gaps are small, so no single category carries the business.

Stores, staff and days

- Store 1 earns 36,997.53 (54.9%). Store 2 earns 30,409.03 (45.1%).
- Staff workload is balanced: Mike Hillyer handled 8,054 payments, only slightly more than the other staff member.
- Tuesday is the busiest day (2,463 rentals) and Thursday the quietest (2,200).
- PG-13 films earn the most (15,259.16). G films earn the least (11,664.23).

4. Recommendations

1. **Prepare for the summer peak.**
    - July and August carry most of the revenue.
    - Stock more copies and schedule more staff then.
    - Try promotions in May and June to build demand earlier.
2. **Deal with the 42 never-rented films.**
    - Check first whether they have any copies in inventory.
    - Stop buying more of them.
    - Discount, move or remove the rest.
3. **Buy deeper on proven titles.**
    - The top 10 films are rented about twice as often as an average film.

    - Add copies and display them prominently.
4. **Look after valuable customers.**
    - Offer a loyalty reward to the 285 above-average customers.
    - Send reminders or small discounts to the other 314.
5. **Close the gap between the stores.**
    - Store 1 earns about 10 percentage points more of the revenue.
    - Compare stock, staffing and location.
    - Move popular titles to the weaker store.
6. **Lift the quiet days.**
    - Test a midweek offer on Thursday.
    - Restock before busy Tuesdays.

5. Limits of this analysis

- Only five months of data, with a long gap in between.
    - Seasonal findings are hints, not proof.
- Revenue is counted in the month of the rental.
- Store figures use the customer's home store, not the store holding the copy.
- No cost data, so these are revenue results, not profit.
- Before acting, collect the missing months, add costs, and test one or two ideas on a small scale.

6. Summary

- The business earns most of its money in a short summer window.
- A small set of popular films and loyal customers drives it.
- A few dozen titles earn nothing.
- The clearest ways to earn more: plan stock better, keep good customers, and close the gap between the stores.